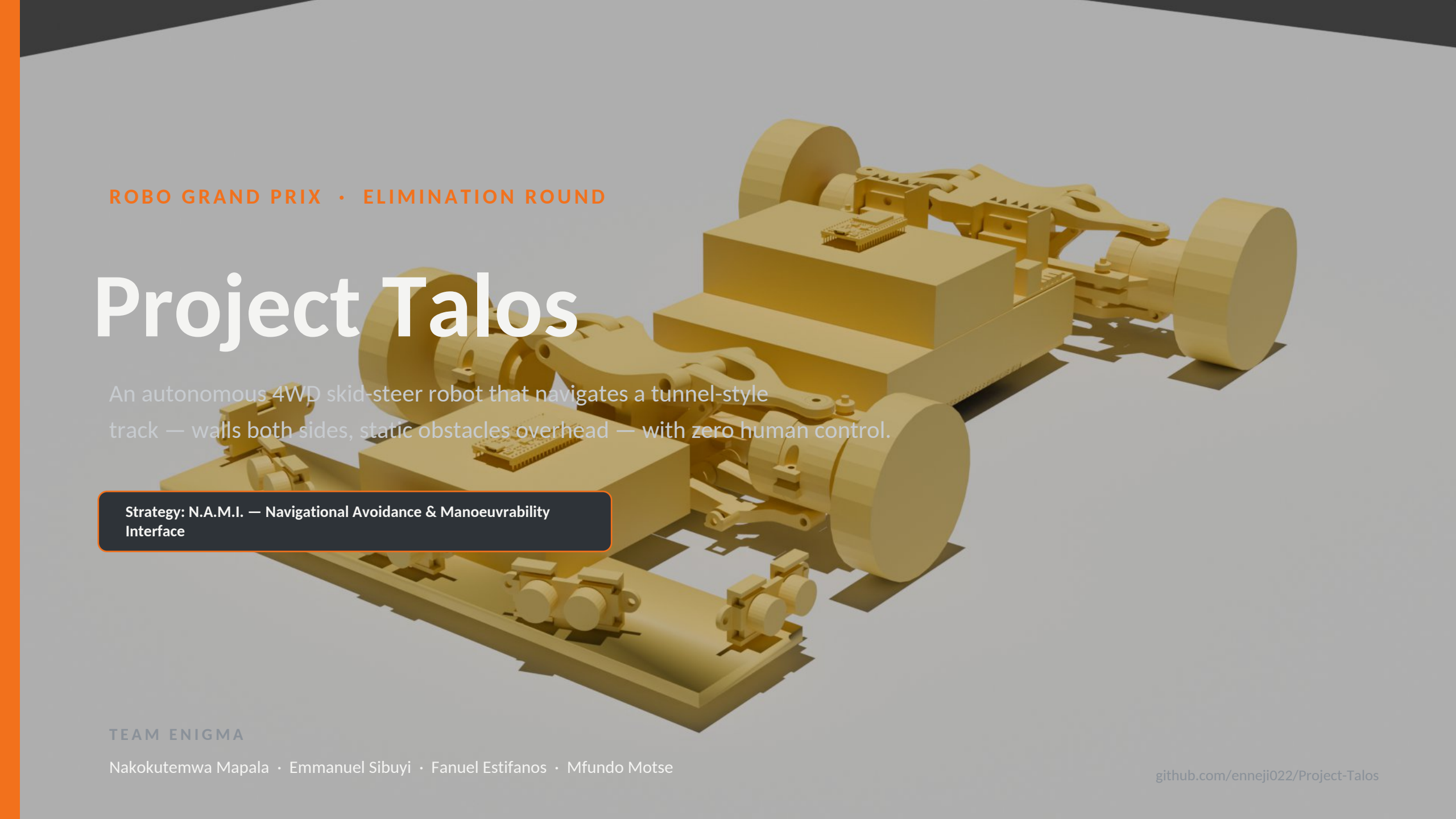




ROBO GRAND PRIX • ELIMINATION ROUND

Project Talos

An autonomous 4WD skid-steer robot that navigates a tunnel-style track — walls both sides, static obstacles overhead — with zero human control.

Strategy: N.A.M.I. — Navigational Avoidance & Manoeuvrability Interface

TEAM ENIGMA

Nakokutemwa Mapala • Emmanuel Sibuyi • Fanuel Estifanos • Mfundo Motse

github.com/enneji022/Project-Talos

Fully autonomous navigation, zero margin for human help

1

Fully autonomous

No remote control, no Wi-Fi link, no human contact once the run starts — any intervention forfeits the run.

2

Tunnel-style track

Walls on both sides plus static obstacles overhead, so the robot must judge open space on the fly, not follow a fixed line.

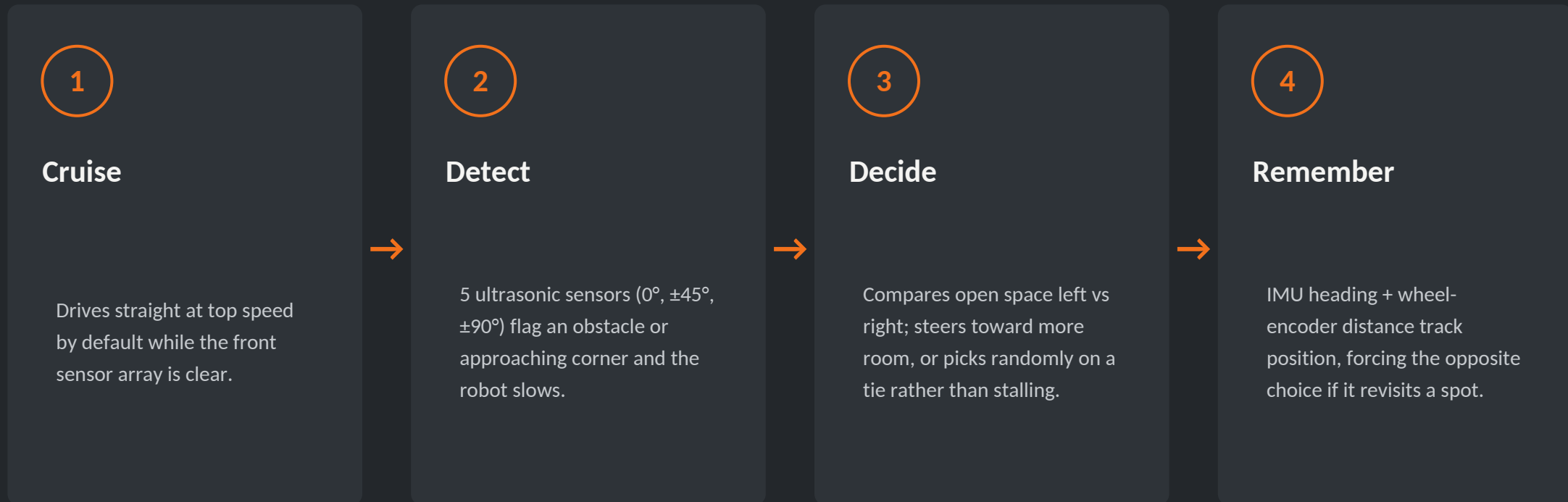
3

Hard size & safety limits

50 × 50 cm footprint, sub-5 kg weight, and a mandatory dual E-Stop (physical + wireless) annotated in the design.

The competition penalty for getting this wrong is total: one human touch and the run is void — so the build has to trust its own sensing completely.

N.A.M.I. — the decision loop that gets Talos through the tunnel



Split across two ESP32 boards: a Sensor board runs all NAMI decision logic; a Motor board only executes drive commands — keeping safety-critical motor control isolated from decision logic.

Two boards, one job split cleanly in two

CORE HARDWARE

- 4WD skid-steer chassis · 4× JGB37-520 geared DC motors w/ encoders
- 2× L298N motor drivers (one per differential side)
- 5× HC-SR04 ultrasonic array (front, near-180° coverage)
- 1–2× IR proximity sensors (backup obstacle detection)
- MPU6050 IMU for heading / position tracking
- 2× ESP32 (sensor board + motor board) linked by UART
- Dual E-Stop — physical mushroom switch + RF kill switch — on a shared cutoff relay

SENSOR BOARD

Runs all NAMI decision logic — reads the sensor array + IMU, chooses steer direction, and sends drive commands over UART.

Build: 6.8% RAM · 23.8% flash used (PlatformIO, esp32doit-devkit-v1) — zero warnings

MOTOR BOARD

Executes whatever drive command it's sent — keeps safety-critical motor control simple and isolated from decision logic.

Build: 6.7% RAM · 21.3% flash used — version-guarded LEDC macro fixes a real ESP32 3.x/older-toolchain API mismatch

TESTING & VERIFICATION

Proven in simulation before a wheel turns

5

Ultrasonic sensors
simulated in Wokwi

0

Compiler warnings
across both boards (Know)

2

Independent PlatformIO
builds verified

WHAT WAS DEMONSTRATED

- Firmware run in the Wokwi simulator against a 5× ultrasonic + MPU6050 + IR rig
- Confirmed CRUISE → DECISION transition when an obstacle enters range, steering toward the side with more space

HONEST LIMITS

- Tie-break coin-flip & loop-prevention override are code-reviewed against spec, not yet individually triggered in a test run
- Board-to-board UART link verified by direct struct byte-matching, not a live combined simulation (no Wokwi part for the TB6612FNG)

TOTAL COST OF IMPLEMENTATION

Budget-conscious build, full BOM traceability

TOTAL (ZAR)

R 2,640

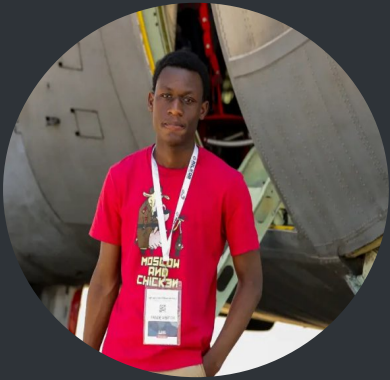
~R500 confirmed live on-site · ~R2,140 estimated,
to be re-checked in-cart before ordering

Full itemised Bill of Materials — component, qty,
unit price, supplier, stock code, direct link — is in
Folder C of the repo.

Category	Item	Qty	Line Total (ZAR)
Compute	ESP32 Dev Board (WROOM-32)	2	R 196
Motor Control	L298N Dual H-Bridge Driver	2	R 72
Sensing	HC-SR04 Ultrasonic (5) + MPU6050 + IR (2)	8	R 360
Drive	12V Gear Motors + Wheels	4 + 4	R 940
Safety	Mushroom E-Stop + RF Kill Switch	2	R 362
Power	12V Battery Packs	2	R 560
Misc	Wiring, connectors, mounting hardware	1	R 150

TEAM

Team Enigma



Nakokutemwa Mapala

Lead • Background Research • 3D
Modelling • Coding



Emmanuel Sibuyi

Engineering



Fanuel Estifanos

Assistant Researcher • Coding



Mfundo Motse

Diagrams • Slide Deck • Technical
Report • Documentation Lead